VIJAYA KARTHIK

Generative AI Engineer
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PROFESSIONAL SUMMARY

Generative AI Engineer who designs, builds and deploys LLM applications. Hands-on with multimodal ingestion, hybrid retrieval, self-hosted model serving, guardrails, evaluation and AWS deployment. Built MAGIK, a multimodal RAG assistant that gives source-cited answers from documents, images, audio and video using open-source models.

TECHNICAL SKILLS

Programming & Backend: Python, REST API, Authentication

Generative AI: LLMs, RAG, Prompt Engineering, Hybrid Search, Reranking, Self-Hosted LLM Inference

Multimodal AI: Computer Vision, OCR, Speech Recognition, Vision-Language Models, Document Parsing

Databases: Vector Databases, NoSQL Databases, Caching

Evaluation & Safety: LLM Evaluation, Hallucination Detection, LLM Guardrails, PII Protection

Cloud & MLOps: AWS Cloud Deployment, Containerization, CI/CD, MLOps, Observability, Experiment Tracking

PROJECT

Multimodal Agentic RAG Integrated Knowledge AI Assistant (MAGIK) v1.0.4 Mar 2026 - Oct 2026

GitHub: github.com/vjkarthik98/MULTIMODAL-AGENTIC-RAG-INTEGRATED-KNOWLEDGE-AI-ASSISTANT

A self-hosted multimodal RAG assistant that answers questions from PDF, Word, Excel, text, image, audio and video files with source citations, using Hugging Face and other open-source models.

- Built separate pipelines for PDF, Word, Excel, text, image, audio and video files using OpenCV, Tesseract, EasyOCR, Whisper and Hugging Face vision-language models on PyTorch, so users can query every format.
- Implemented hybrid search that combines BM25 keyword search with semantic search over BGE embeddings in the Qdrant vector database, then reranks the results with a cross-encoder for the most relevant LLM context.
- Designed an agentic query router that decides whether each question should be answered from the uploaded documents, web search, chat history or directly by the LLM.
- Created a self-verification step that checks numbers, citations and groundedness against the source documents and searches again when an answer is not well supported, to reduce hallucinations.
- Self-hosted a quantized GGUF Qwen2.5-14B model with llama.cpp on a GPU, served through a FastAPI REST API that streams answers with source citations and keeps chat history in Redis and MongoDB.
- Added guardrails that block prompt injection and jailbreak attempts, strip hidden instructions from uploaded files and mask PII with Presidio, plus JWT and Google OAuth 2.0 authentication with per-user data separation.
- Set up LLM evaluation with RAGAS, DeepEval and MLflow experiment tracking, using a local LLM-as-a-Judge, to measure retrieval quality, answer quality and hallucinations.
- Containerized with Docker and deployed on AWS EC2 (GPU) via GitHub Actions CI/CD, with Prometheus, Grafana and OpenTelemetry observability and AWS Lambda functions that stop the idle GPU server and restart it on demand.

CERTIFICATIONS

Google Cloud | Responsible AI for Developers Specialization Mar 2026

Duke University | Machine Learning Operations (MLOps) Specialization Mar 2026

IBM | Generative AI Engineering Professional Certificate Nov 2025

PERSONAL PURSUITS

UPSC Civil Services Examination Preparation 2021 - 2025

EDUCATION

Master of Arts (English) | Bishop Heber College, Tiruchirappalli 2019 - 2021

Bachelor of Arts (English) | Don Bosco College, Dharmapuri 2016 - 2019